Portable Cooling Shelter Current Design Plan

Calculated Visions
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Current baseline as of October 11 2026

Design direction

We are developing a compact, collapsible, one-person cooling shelter for outdoor crews working in remote areas. The shelter combines a reflective insulated enclosure with an end-mounted conventional compressor AC module. It is a plug-in product using the crew's existing generator or a compatible external electrical supply. The purpose is to bring a cooled break location closer to workers when returning to a truck, trailer or building is inconvenient.

The current priority is a full-size fit mockup followed by a functional cooling prototype. Keep the refrigeration system factory sealed for the first working build, and develop the frame, shell, detachable interface, airflow, drainage and controls around measured cooling performance. The current record establishes design work and a scale model; it does not establish full-size thermal, electrical or field performance. [1–5]

Design element	Current baseline
Use and capacity	One worker taking a short cooling break; remote construction, forestry, conservation and wildfire-support use.
Shelter geometry	6.5 ft long × 2.5 ft wide × 40 in peak height; 16.25 ft² floor area. Retain pending confirmation against current CAD.
Structure and entry	Rectangular base, curved roof, two diagonal crossed poles and side entrance.
Envelope	Reflective appearance, weather-resistant exterior and thin insulation; exact layer stack remains open.
Cooling and power	Conventional vapor compression, exterior end module, separated hot and cold air circuits, plug-in power.
Completed work	Fusion CAD and a 1:12 physical model are recorded in the latest presentation. Full-size validation is next.

How to read this plan

Baseline decisions describe the team's current direction. Proposed targets are engineering goals that require tests. Estimates support sizing and budget decisions; they are not achieved specifications. Recommendations in this plan remain open for team approval. The team is a student project group, not a MESA corporation.

Shelter geometry and mechanical design

Preserve the low-profile arched enclosure as the starting geometry. The small floor area limits material and conditioned space, while the crossed poles are intended to support the roof and resist movement in more than one direction. Wind resistance and stability still need measurement. [1,2]

Feature	Design action
Frame	Two flexible poles run corner to opposite corner and cross above the base. Select pole sections and attachment points after a full-size assembly trial.
Side entrance	Retain an arched side opening and roll-up concept. Final opening width, height, threshold and zipper arrangement are not frozen.
AC end connection	Create a short supply and return interface through the slanted end region. A flange or collar must seal to the shell without carrying module weight.
Module support	Support the AC on the ground or a dedicated frame cradle. Add vibration isolation and allow service access.
Floor	Use a cleanable moisture barrier with thin insulation. Confirm that an occupant can sit or recline without discomfort from uneven ground.
Anchoring and packing	Provide stakes or suitable alternative anchors, protected pole ends and a carry arrangement. Measure the complete packed system.

Resolve posture before freezing dimensions

At 40 in peak height, usable head clearance is lower away from the roof center. A person may fit lying down while lacking comfortable seated clearance. Test representative adults in work clothing using a full-size inexpensive mockup. Check entry, normal sitting posture, reclined posture, turning and immediate unassisted exit. Enlarge the local roof or door if needed; then update cooling load, material area and packing. California's outdoor shade provision specifically requires normal seated posture for resting workers. [6]

Volume and drawing control

The rectangular volume bound is 54.2 ft^3. An ideal half-elliptical arch prism gives approximately 42.5 ft^3, but the actual end slopes and crossed-pole shape can reduce volume. Export enclosed volume, surface area, door clear opening, module bounding box and interface dimensions from CAD. Do not size AC from floor area or air volume alone. [3]

Appearance and hardware

Retain the reflective shelter appearance and orange industrial module concept. The CAD includes interior and exterior fan or vent representations; their exact count and position must follow the selected heat exchangers and air paths. Place hard projections away from a person's back during carry. Add coil guards and grilles while preserving adequate airflow. A visualization texture does not establish the material's solar or thermal properties.

Envelope and insulation plan

Use a durable exterior skin plus a removable thin reflective foam liner as the first testable construction. Choose the exterior for sunlight rejection, weather exposure and repeated folding. Choose the liner for conduction resistance, radiant exchange and packability. Treat exterior solar reflectance and foil thermal emittance as separate properties. [2]

Layer or detail	Proposed construction and verification
Exterior	Weather-resistant textile or coated skin with documented solar reflectance, cleanability and UV suitability. Confirm fabric and seam requirements for the intended use.
Insulating liner	Compare Low E Micro E about 3.2 mm, Prodex about 5 mm and Reflectix double bubble about 7.9 mm using samples. These are research candidates, not selected production tent materials.
Reflective air space	Trial a 10–20 mm separation where low-emittance foil faces an air space. Keep any ventilated outer cavity separate from conditioned interior air.
Joints and openings	Minimize leakage at zippers, floor joins and the AC flange. Use repairable seals that tolerate folding and assembly.
Moisture	Provide removable liner access, drainage and drying. Protect wiring from conductive foil and prevent water collecting in inaccessible layers.

Radiative coating remains an experimental option

Cryo X Cool coating was previously approved for prototype investigation. It is not yet an established layer of the final assembly. Test adhesion to the chosen flexible substrate, folding damage, mass, moisture resistance and temperature benefit against an uncoated control. Do not assume that a coating and a shiny foil surface perform the same thermal function.

Material comparison procedure

Use interchangeable panels or matched enclosures with the same exposed area, orientation, openings and sensor positions. Compare ordinary fabric, candidate liners and the preferred liner with an air gap. Record ambient and interior air temperature, contact surface temperatures, humidity, wind and sun conditions. Repeat side-by-side tests and swap positions. In AC trials, hold equipment and setpoint constant and compare energy use. [2]

Building assembly R values cannot be assigned to a thin tent liner. A shiny finish alone does not prove low solar heat gain. Obtain SKU-specific data and validate the assembled shell; quarter-scale material tests cannot establish full-size cooling performance. [2]

Compressor AC module design

Retain conventional vapor compression. The sealed refrigerant path is compressor → condenser → expansion device → evaporator → compressor. The evaporator removes heat and moisture from the occupied space; the condenser releases heat outdoors. A variable-speed DC compressor remains a conventional compressor system. [3]

Assembly	Integration requirement
Compressor and driver	Supplier-matched sealed refrigeration core, relevant capacity map, vibration isolation and restart protection.
Evaporator and blower	Insulated cold plenum, useful supply distribution and representative return-air sensing.
Condenser and outdoor fan	Unobstructed outdoor intake and exhaust; reject cooling load plus electrical input heat.
Expansion device and circuit	Retain the supplier's matched charge, valve or capillary, lines and refrigerant. Do not select loose components solely by advertised wattage.
Condensate system	Drain pan and outlet away from floor, wiring and walking routes; address drain blockage.
Controls and protection	Thermostat, coil-temperature sensing, anti-ice logic, overtemperature protection and visible cooling fault indication.

Keep hot and cold air paths separate

Cold circuit: shelter return air passes through the evaporator and returns as cooled supply air. Hot circuit: outdoor air passes through the condenser and exhausts outdoors. Seal the partition between circuits. Locate supply and return far enough apart to avoid immediate short-circuiting. The existing CAD's fan openings are placeholders until these paths are defined. [3]

Use short insulated supply and return passages, initially preferably under 0.5 m each. Start duct exploration around 100 –125 mm internal diameter; confirm installed flow and pressure losses before fixing flange size. Proposed cold airflow is 60–100 CFM. Free-air fan ratings are insufficient after coils, ducts and guards are installed. [3]

Prototype procurement route

An intact portable compressor AC can validate the full-size shell before a compact OEM package is customized. The October 10 research identifies RIGID micro-AC modules and ZERO BREEZE or EcoFlow portable ACs as screening benchmarks. No unit is selected. Request capacity at actual return and ambient conditions, sensible and total cooling, installed airflow, input demand, permitted operating temperature and dimensional drawings. Reconfirm availability and quotes before purchasing. [3]

Cooling capacity and plug in power

Use the preliminary thermal load model to choose a test module, then replace assumptions with measurements. The load includes shell conduction, solar gain, floor heat, the worker, ventilation, leakage and equipment. Cooling output watts and electrical input watts are different quantities. [3]

Scenario from prior research	Continuous load	With 20 percent allowance
Controlled shade at 35°C ambient and 25°C interior	540 W	648 W
Design case at 35°C ambient and 25°C interior	750 W	900 W
Severe case at 43°C ambient and 26°C interior	1,286 W	1,543 W

These are screening estimates, not measurements. The design case assumes about 6 m² shell area, effective U = 5 W/m²K, 200 W occupant load and a provisional outside-air load. Its 5 CFM ventilation assumption is not an approved occupied ventilation rate. Determine fresh-air needs separately and add the resulting thermal load. Avoid double-counting solar gain when measured hot-surface temperatures are used. [3]

Preliminary module target

Target 900–1,200 W total cooling, approximately 3,100–4,100 BTU/h, including 600–800 W sensible cooling at stated conditions. A 900 W catalog rating leaves little margin above the estimated 750 W design load once hot-weather derating and installation losses occur. Severe conditions may need a 1,500–2,000 W class module, more shade or a narrower operating envelope. [3]

External power architecture

Proposed U.S. input: 120 V, 60 Hz from a compatible crew-owned generator, using the selected AC's approved input or approved mains-to-DC supply. Final connector, voltage, current, grounding and fault protection remain to be specified. Internal 24 V or 48 V operation may suit an OEM compressor without changing the user-facing plug-in product. [3,5]

Reserve an initial 300–500 W running electrical budget and measure startup peaks, power factor and maximum draw. At 120 V this is nominally 2.5–4.2 A only at unity power factor. Size the input and extension cord from actual ratings, cord length and applicable installation requirements. At 400 W average draw, eight hours uses 3.2 kWh; this is planning arithmetic. [3–5]

No generator, battery or solar package is included in the baseline product or cost. Remote use still requires a compatible power source within practical safe cable reach. The product can be carried beyond road access only where that external-power requirement can also be met.

Proposed requirements and design gates

The following values are proposed acceptance targets. None should be described as achieved until the test record confirms it. [3–5]

Requirement	Proposed target or release gate
Capacity and posture	One adult; comfortable intended rest posture and normal sitting clearance assessed in a full-size mockup; immediate unassisted exit.
Occupied-zone cooling	At 35°C ambient, target 25–27°C occupied air with a reproducible 200 W simulated occupant load and declared sun, humidity, ventilation and opening conditions.
Cooldown	Proposed minimum 5°C occupied-zone reduction within 10 minutes after defined heat soak; record 5, 10 and 15 minute values and steady conditions.
Thermal envelope	Test hotter conditions within supplier limits. Define required shade and maximum ambient from measured performance.
Module mass and size	Aspirational finished AC module 8–12 kg and 15–25 L. Allow a larger first test unit to establish function.
Complete carry package	Provisional total 15–22 kg including shelter, module, adapter, passages, cable and bags. Generator remains separate; define and test the intended carry task.
Deployment	Proposed one-person setup within 3 minutes without tools, including power, duct connection and anchors; identify any cooling wait separately.
Noise	Proposed interior target ≤55 dBA under stated mode and measurement position; revise after prototype assessment.
Ventilation and water	Deliberate fresh-air exchange; safe occupied review; no condensate into floor or electrics and no outdoor exhaust recirculation.
Durability	Repeated fold, setup, carry, anchoring and cleaning trials; select cycle count and environmental limits before testing.
Failures	Clear loss-of-cooling indication, quick-open exit and a planned fallback rest location; unoccupied power-loss heat-rise test.

Resolve conflicting goals with evidence

If the target cannot be met in direct sun, establish a shade requirement or increase capacity and revise mass and cost. If 40 in height compromises sitting or exit, enlarge the shelter and re-run load estimates. If total carry mass is too high, separate shelter and AC into two loads and validate two-person transport. Do not reduce condenser clearance, drainage or fresh-air provision simply to fit the present CAD box.

Prototype build and validation sequence

Stage	Work and evidence needed before proceeding
1 Full-size fit mockup	Build the current footprint and roof with low-cost materials. Measure occupied clearance, doorway, exit and module placement. Freeze posture and revise CAD if needed.
2 Materials and bench cooling	Compare liner samples. Bench-test a factory-sealed AC with the intended supply, return, guards and drain. Record cooling conditions, airflow, power and faults.
3 Full-size integration	Join the frame, shell, insulated floor and AC interface. Check mechanical support, leakage, drainage, protection and cable routing.
4 Unoccupied thermal validation	Use a 200 W heat-load surrogate; separately account for moisture. Compare shell variants and the ordinary-tent benchmark under documented conditions.
5 Failure and relocation trials	Measure heat rise after loss of power, easy opening, blocked-drain response within safe equipment limits, packing, complete mass, setup and terrain carry.
6 Supervised user feedback	After safe operation is reviewed, obtain comfort, posture, access, observation and practical field feedback. No deliberate heat-illness or extreme-heat challenge.

Thermal test record

Log ambient temperature and humidity, sun or shade, wind, interior air at multiple occupied positions, supply and return temperatures, contact shell temperatures, electrical power and energy, drainage and opening times. Protect air sensors from direct sun and cold outlet jets. Begin from repeatable heat-soaked conditions. Report 5, 10 and 15 minute results plus stable performance; a cold outlet alone is insufficient.

Repeat comparable tests at least three times and identify weather differences. Start with the design condition, then assess hotter conditions within equipment limits. A dry electrical heater represents sensible heat only; it does not reproduce the worker's full latent moisture load. Add a controlled moisture test after bench drainage checks. No one should occupy a heat-soaked test enclosure to establish its safety.

Comparisons and pass decisions

Compare the integrated design against a small ordinary tent using the same AC, power and ventilation assumptions. Assess temperature, humidity, energy, mass, packed volume and setup. Verify that improving the shell reduces installed cooling demand enough to justify its weight and cost. Keep the 1:12 model for geometry and communication; it cannot establish full-size cooling capacity.

Field deployment and commercial plan

The strongest initial customer is an employer supporting remote crews with available electrical power but inconvenient access to existing cooled spaces. Forestry, conservation, remote construction and wildfire-support interviews support the access problem. The latest presentation records 6+ interviews; interview recollections describe needs rather than measured product outcomes. [1]

Heat safety fit

OSHA guidance identifies cooler rest locations, including air-conditioned spaces and tents. California §3395 requires adequate nearby shade, normal seated posture, sufficient concurrent capacity and monitoring during preventative rests. One-person throughput does not establish enough capacity for a crew. The product can support the employer's heat plan, but it does not replace hydration, training, observation or emergency procedures. [6,7]

Make the occupant visible or readily checkable, allow communication and preserve rescue access. Provide a cooling-failure indication and quick-open enclosure. Keep generator exhaust away from the tent and fresh-air inlet according to the generator's instructions and competent placement review; keep cords and plugs protected from water and trip hazards. The shelter is not a validated smoke refuge or stand-alone heatstroke treatment system. [5]

Differentiation to prove

The proposed advantage is a compact, collapsible, purpose-built remote-worker cooling station. A rigid booth such as TRUSCO uses freight and handling logistics; a trailer requires road access. A small tent paired with portable compressor AC is the closest practical comparison and may also serve remote sites. Prove the integrated shelter's advantage through complete carry mass, packed volume, setup, cooling and energy measurements rather than claiming no competing solution exists. [4]

Preliminary end user cost

October 10 planning estimate	Amount and interpretation
Central manufacturing allowance	$1,630 per complete plug-in shelter and AC system.
Central direct-sale price	Approximately $2,500, using cost ÷ 0.65 for 35% gross margin.
Early direct-sale range	$2,000–$3,000; not a quote or final MSRP.
Included allowances	Refrigeration core, frame and textile, liner, case and seals, electrical interface and controls, assembly, testing, packaging and warranty reserve.
External and unresolved costs	Crew-owned generator excluded. Taxes, delivery and nonrecurring development, tooling and product evaluation require separate budgeting.

The higher-capacity refrigeration core is not yet quoted. A larger AC or taller enclosure can raise price and carry mass. Obtain three refrigeration quotations and two enclosure quotations, then update the bill of materials from a dimensioned drawing and test configuration. [4]

Decisions still open

Confirm whether the current CAD retains the 6.5 × 2.5 ft footprint and 40 in peak height; finalize rest posture and doorway; choose liner and exterior; select AC and module size; define input and cable limits; establish fresh-air design; set the tested ambient envelope; and measure complete carry mass. These decisions should precede production drawings or a firm selling price.

Project references

[1] Calculated Visions MESA Human Centered Final presentation, October 10–11 2026, especially design, interview and validation slides. Earlier team CAD discussions supply numeric geometry, crossed poles, side entry and orange module direction.

[2] MESA Reflective Insulation Research, October 10 2026. Material screening and proposed shell tests.
[3] MESA Compressor AC Module Research, October 10 2026. Load scenarios, commercial benchmarks and proposed AC targets.
[4] MESA Market Competition and Cost Report, October 10 2026. Revised remote-worker positioning, plug-in scope and cost model.
[5] MESA OSHA Safety and Design Fit, October 10 2026. Employer integration and proposed product checks.

[6] California Department of Industrial Relations, Title 8 Section 3395, especially shade and rest provisions. Checked October 11 2026.
https://www.dir.ca.gov/title8/3395.html
[7] OSHA, Heat Water Rest Shade guidance. Checked October 11 2026.
https://www.osha.gov/heat-exposure/water-rest-shade

Commercial specifications and costs above are the October 10 research snapshot. New targets in this plan are proposals; quotes, supplier suitability, complete-product review and measured prototype performance remain to be established.